



Molycorp (A): Why this case?

Financial Strategy XP 35802
Winter 2026

Per Strömberg

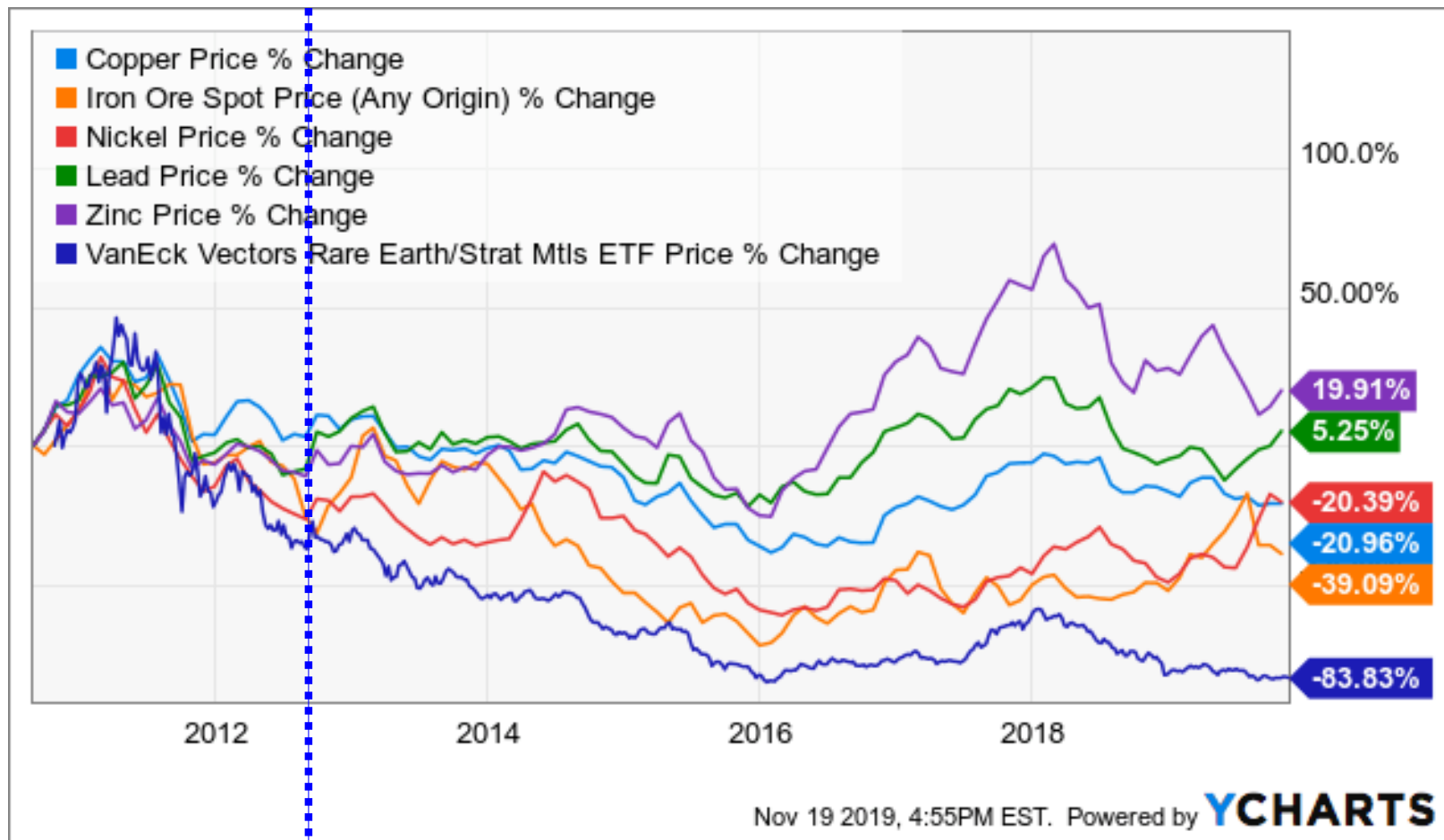
University of Chicago
Booth School of Business



What Happened?

- August 16, 2012: Announced intention to raise \$450 million
 - Resized offering to \$480 million (net proceeds of \$457 million)
 - » Convertible Notes: \$360 million (6% coupon)
 - » Primary Shares: \$120 million (\$10/share)
- MCP Stock Price Reaction
 - August 16th: Down 7.5% to \$11.16
 - August 17th: Down 11.8% to \$9.84
- Commenting on the Deal
 - CIBC Analyst: *“We believe that this gives the company plenty of cushion to absorb any bumps in the road during the ramp up phase at Mountain Pass”*
 - JP Morgan Analyst: *“We remain cautious on the stock as we still don’t fully know MCP’s exact cash requirements over the next year”*

REO Prices Continued to Fall



Molycorp Stock Price Performance



Subsequent Events

■ Cash flow from operations:

- 2012: (\$90) million
- 2013: (\$154) million
- 2014: (\$451) million

■ Additional Financings:

- 1/23/2013: \$300 million
 - » \$100 million of senior secured convertible notes @ 5.5%
 - » \$200 million of common Stock (\$6/share):
- 10/15/2013: \$200 million of common stock (\$5/share)

■ Leverage:

- 2012: 52%
- 2013: 54%
- 2014: **87%**

Stock Price:

\$9.44
\$5.62
\$0.88

■ Bloomberg News reports:

- July 9, 2014: Apollo Management has acquired a substantial position in MCP convertible notes in order to gain control of the rare earth miner if it defaults on its debt.
 - » MCP stock price falls 17% to \$1.88/share
- March 18, 2015: Apollo in “restructuring talks” with MCP
 - » Owns \$132 million of \$230 million convertible debt due June 2016
 - » MCP stock price closes at \$0.37/share

■ April 15, 2015: Molycorp chosen to supply rare earths for use in high-efficiency Siemens wind turbine generators

- MCP stock price increases 75% from \$0.54/share to \$0.94/share

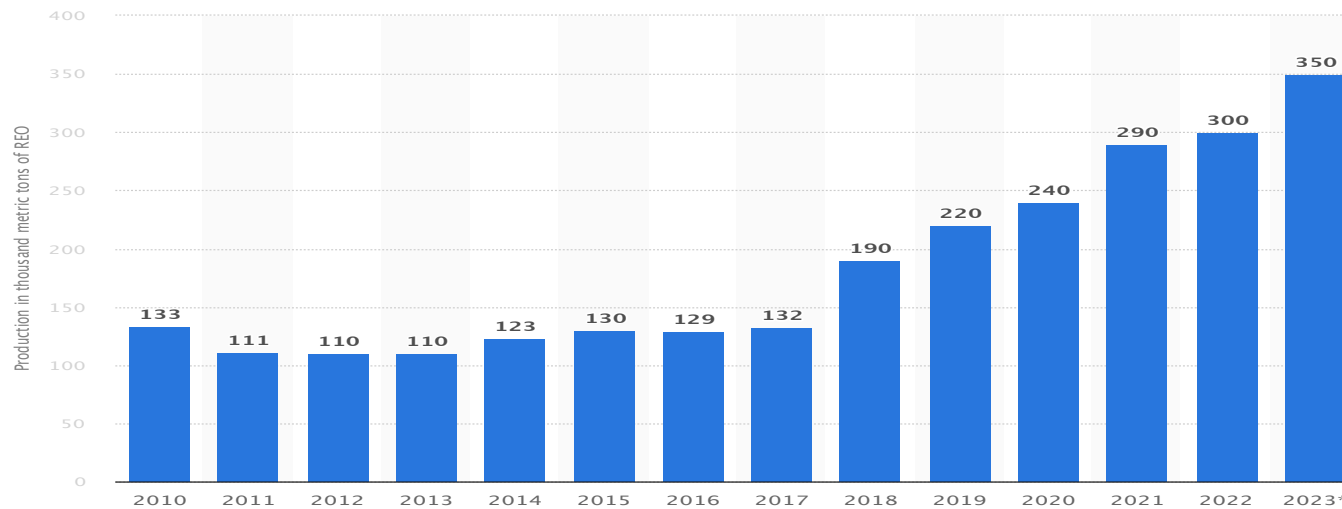
■ **Project Phoenix: Rising from the ashes?**

■ June, 2015: files for bankruptcy

- At Ch. 11 filing: \$1.7 billion of debt, book assets \$2.1 billion, \$21 million of available cash
 - China's relaxed export rules
 - Battery technologies using less of rare earth metals
 - Demand (largely in China) dropped off faster than expected
 - Processing facility ramping up output slower than expected
- Reorganization plan approved May, 2016
 - Assets went to distressed investors holding Molycorp debt
 - Mountain Pass mine to hedge funds JHL Capital & QVT Financial and Chinese mining firm Shenghe (8% non-voting stake). Renamed "MP Materials Corp"
 - » Went public on NYSE in 2020 through reverse merger
 - Rest of company to distressed investor Oaktree. Renamed "Neo Performance Materials"
 - » IPO 2017 on Toronto Stock Exchange

Mine production of rare earth elements worldwide from 2010 to 2023

(in 1,000 metric tons REO)



VanEck Rare Earth and Strategic Metals ETF

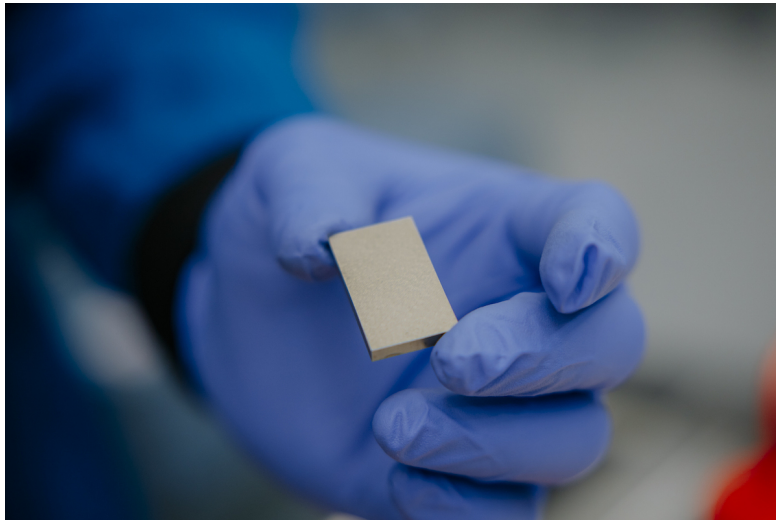
NYSEARCA: REMX

400 247.32 USD 29 Oct 2010



MP Materials Restores U.S. Rare Earth Magnet Production

01/22/2025



[Download this image](#)

Commercial NdPr Metal Production Begins at Independence, MP's Fully Integrated Rare Earth Magnet Manufacturing Facility in Texas for the First Time in a Generation

Trial Production of Automotive-Grade, Sintered NdFeB Magnets Underway, with First Deliveries on Track for Year-End



"This milestone marks a major step forward in restoring a fully integrated rare earth magnet supply chain in the United States," said James Litinsky, Founder, Chairman, and CEO of MP Materials. "With record-setting upstream and midstream production at Mountain Pass and both metal and magnet production underway at Independence, we have reached a significant turning point for MP and U.S. competitiveness in a vital sector."

NdFeB magnets — the world's most powerful and efficient permanent magnets — are essential components in vehicles, drones, robotics, electronics, and aerospace and defense systems. Despite their indispensable role, the U.S. has relied almost entirely on foreign sources for these critical inputs for decades.

MP Materials is addressing this gap by building America's first fully integrated rare earth metal, alloy, and magnet manufacturing facility. With commercial NdPr metal production already online and trial production of sintered magnets underway, Independence is poised to produce approximately 1,000 metric tons of finished NdFeB magnets per year, with a gradual production ramp beginning in late 2025. The facility will supply magnets to General Motors and other manufacturers, sourcing its raw materials from Mountain Pass, MP Materials' mine and processing facility in California.

In 2024, MP Materials achieved record-breaking production at Mountain Pass, delivering more than 45,000 metric tons of rare earth oxides (REO) contained in concentrate — an all-time high for U.S. primary production. Mountain Pass, America's only active rare earth mining and processing operation of scale, also set a midstream production record, producing approximately 1,300 metric tons of NdPr oxide in 2024, in addition to cerium, lanthanum, and other separated and refined products.

Ukraine's rare earth elements could help keep US military aid flowing



BY **SAMYA KULLAB**

Updated 1:35 AM GMT+8, February 12, 2025



KYIV, Ukraine (AP) — [Ukraine](#) has offered to strike a deal with U.S. President Donald Trump for continued [American military aid](#) in exchange for developing Ukraine's mineral industry, which could provide a valuable source of the rare earth elements that are essential for many kinds of technology.

Trump said that he [wanted such a deal](#) earlier this month, and it was initially proposed last fall by Ukrainian President Volodymyr Zelenskyy as part of his plan to strengthen Kyiv's hand in future negotiations with Moscow.

"We really have this big potential in the territory which we control," Andrii Yermak, chief of staff to the Ukrainian president, said in an exclusive interview with The Associated Press. "We are interested to work, to develop, [with our partners](#), first of all, with the United States."

<https://apnews.com/article/russia-ukraine-war-rare-earth-elements-trump-6a4a95d4370a2e46a81ce4a3a352bf8f>

Molycorp - Why this case?

- Another comprehensive capital structure analysis from start to finish
- Example of how to use valuation to help make capital structure decisions
 - Assess and account for mispricing
 - Account for financial distress and equity option value
- Introduction to valuation

Steps in making a capital structure decision

- (1) Understand the the business and economic environment
 - Qualitatively and quantitatively (historical ratios, S&U)
- (2) Qualitatively evaluate net benefits of debt
 - $PVTS + PV(Inc.) - PV(COFD)$
- (3) Pin these insights down to a target leverage ratio
 - Look at ratios for comparable companies
 - Look at debt ratings
 - Estimate future cash needs in a cash flow statement (S & U)
- (4) Consider dynamic tradeoffs
 - Any reasons for temporarily deviating?
 - New in Molycorp: use valuation to assess misvaluation
- (5) Analyze policies in scenario analysis
 - Focus on being able to handle bad case scenario. “Stress test”

Static Tradeoff for Molycorp:

- P(FD)? High.

1. Volatile prices due to geopolitical risk, new capacity added with a lag
2. Implementation risk in Mountain Pass / Phoenix project.

- COFD:

1. Need capital to finish MP project.
2. Need capital to undertake more acquisitions?

Key question: what is cost of delaying project?

Is this really positive NPV investment, given low prices?

Adding MP capacity likely will push price even lower?

■ PVTs?

- Low benefits currently due to past loss carryforwards (NOLs).

■ Incentive benefits? Depends heavily on view of management's strategy, and value of Project Phoenix

- This is a great investment opportunity → incentive benefits low
- This project is questionable (management does not believe in it, not great track record in executing, better wait until prices increase) → incentive benefits of keeping management on a “tight leash” are high, do not want to give them too much financial flexibility

■ Trade-off analysis ⇒ Depends on your view of project

- Good investment opportunity → low leverage, more financial flexibility good
- Not obviously good investment opportunity → high leverage could be efficient to keep up management discipline

Quantitatively

■ Comps?

- Most have lower leverage → indication Molycorp is overleveraged
- Caveat: Molycorp is vertically integrated, while most comps just do exploration and mining
 - » Exception is Lynas, which also has higher leverage

■ Debt rating target?

- Not clear. Financial plan indicates this round of financing is enough → future access less important.
- Do we really want to make it easy for firm to raise more capital in the future, if current financing is not enough?

Dynamic Issues in Cap Struct. Choice

- Given high leverage, might reach conclusion that company needs equity.
- But how costly is it to adjust to lower leverage target?
- Depends on whether Molycorp is misvalued:
 - Undervalued → issuing equity is expensive.
 - » Convertibles can be an attractive option, like MCI → reach optimal lower-leverage target over time.
 - Overvalued → great timing to issue equity

Using valuation to help make capital structure decision

- Use DCF to assess whether company is under- or overvalued
- Gives explicit value of dilution / undervaluation cost from equity issue
 - Can compare against COFD
 - E.g. value of financial flexibility, value of investments / acquisitions foregone, etc.
- In this case, DCFs indicates that equity might actually be overvalued

Estimating dilution cost of new equity issue for old shareholders:

- $E^f / N^0 = P^f$, fundamental (i.e. true) price per share.
 - N^0 = # of shares outstanding before new equity issue
- If market is not completely inefficient, eventually (3 months? 1 year?) the stock should trade at P^f
- We are issuing N^I new shares at a price $P^I \neq P^f$,
 - Raising $P^I * N^I$ in new cash.
 - I.e. $P^I < P^f \rightarrow$ Equity undervalued, issue will be dilutive
- New equity value after issue = $P^f * N^0 + P^I * N^I$
- New per share value $P^N = (P^f * N^0 + P^I * N^I) / (N^0 + N^I)$
 - $P^I < P^f \rightarrow P^N < P^f$ and existing shareholders lose
 - $P^I > P^f \rightarrow P^N > P^f$ and existing shareholders gain.

■ WACC valuation:

- Get after-tax cash flows to all-equity firm (FCFs)
- Discount at $WACC = R_E * E / (E + D) + R_D * (1 - t) * D / (E + D)$
- E and D are *market* values of the *future* equity and debt
- R_E is return on *levered* equity using *future* E and D.
 - » First, unlever to get $\beta_A = \beta_E^{old} * E / (E + D)^{old} + \beta_D^{old} * D / (E + D)^{old}$
 - » Second, relever to get $\beta_E^{new} = \beta_A + D / E^{new} * (\beta_A - \beta_D^{new})$
 - » Third, calculate $R_E^{new} = R_f + \beta_E^{new} (R_m - R_f)$
 - » Alternatively, calculate $R_E^{new} = R_A + D / E^{new} * (R_A - R_D^{new})$

■ Disadvantage of WACC vs. APV:

- If D/E changes, should recalculate R_{WACC} every year
 - » Circular problem: need valuation to calculate new D/E

■ How to find R_D – alternatives:

(1) Use R_D implied by debt beta (which in turn is implied e.g. by credit rating)

(2) Look at current interest rate on the firm's debt

■ Problem with (2) : R_D is *expected*, not promised interest

– $R_D < \text{coupon rate on debt}$

■ Example: assume a one year loan

– Coupon rate on loan $I = 10\%$, default probability per year 5% , recovery in default $X=60\%$

– $R_D = \text{Prob}(\text{no default}) * I + \text{Prob}(\text{default}) * (X - 1)$
 $= 0.95 * 10\% + 0.05 * (-40\%) = 7.5\%$

– Can get default probabilities for different credit ratings from Moodys or S&P

– Getting recovery rates are harder

» Moodys /S&P have some (proprietary) data; otherwise make assumption that you think is reasonable.